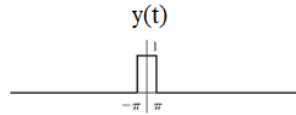
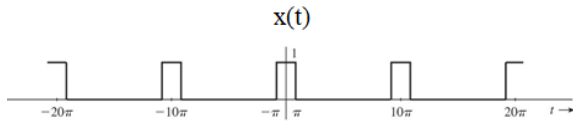


ENGR 3323: Signals and Systems

HW 10_Ch7

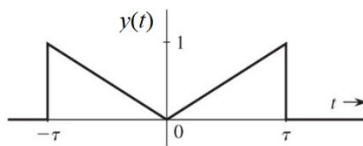
Q1) In previous homework you found the exponential Fourier series of $x(t)$. Find the Fourier transform of $y(t)$ and compare $X(\omega)$ to the Fourier series coefficients D_k .



$$Y(\omega) = 2\pi \operatorname{sinc}(\omega\pi)$$

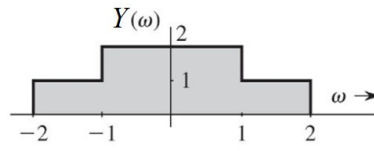
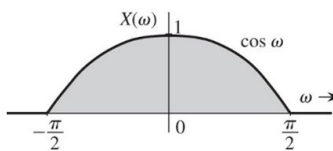
$$D_n = \omega_0 \operatorname{sinc}(n\omega_0\pi)$$

Q2) Find the Fourier transforms of the signals shown below using the Fourier transform integral.



$$X(\omega) = \frac{4 - 2e^{-j\omega} - 2e^{-j2\omega}}{j\omega}; \quad Y(\omega) = \frac{2}{\omega^2} [\cos \omega\tau + \omega\tau \sin \omega\tau - 1]$$

Q3) Find the inverse Fourier transforms of the signals shown below using the inverse Fourier transform integral.



$$X(t) = \frac{1}{\pi(1-t^2)} \cos\left(\frac{\pi t}{2}\right)$$

$$y(t) = \frac{\sin 2t + \sin t}{\pi t}$$

Q4) Sketch the following functions:

a) $\operatorname{rect}(t/2)$

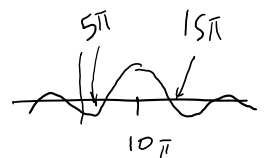
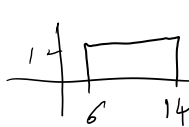
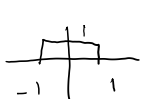
b) $\Delta(3\omega/100)$

c) $\operatorname{rect}((t-6)/8)$

d) $\operatorname{sinc}(\pi\omega/5)$

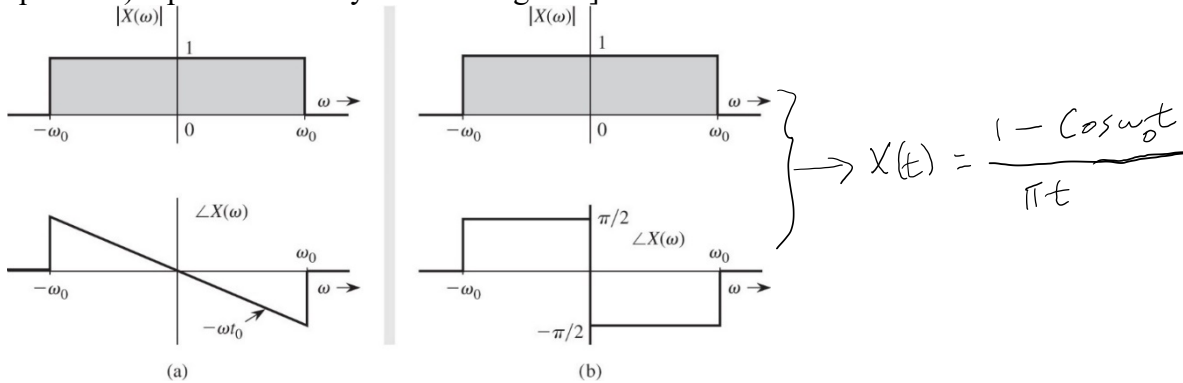
e) $\operatorname{sinc}((\omega/5) - 2\pi)$

f) $\operatorname{sinc}(t/5) \operatorname{rect}(t/10\pi)$



Q5) Find the inverse Fourier transform of $X(\omega)$ for the spectra illustrated in Figure below.

[Hint: $X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$. This problem illustrates how different phase spectra (both with the same amplitude spectrum) represent entirely different signals.]

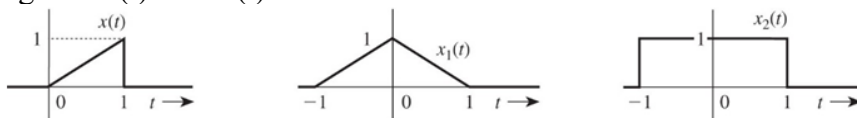


$$X(t) = \frac{\omega_0}{\pi} \text{sinc}[\omega_0(t-t_0)]$$

Q6) The Fourier transform of the triangular pulse $x(t)$ in Figure below is expressed as

$$X(\omega) = \frac{1}{\omega^2} (e^{-j\omega} - j\omega e^{-j\omega} - 1)$$

Use this information, and the time-shifting and time-scaling properties, to find the Fourier transforms of the signals $x_1(t)$ and $x_2(t)$.



$$X_1(\omega) = X(\omega) e^{j\omega} + X(-\omega) e^{-j\omega}$$

$$X_2(\omega) = 2[X(2\omega) e^{j\omega} + X(-2\omega) e^{-j\omega}]$$